

Demo Guide — Quantum Feature Mapping Lab

Presenter's reference for walking a business audience through the demo. Section order matches the app, top to bottom. All numbers are real, unmodified results from the Kipu Quantum Hub benchmark run.

0. Say this first

What the model does: predicts when a hoist cable is heading toward a rupture, early enough to act before it happens.

The core message — repeat it at the start, the middle, and the end:

We do not touch or replace the client's ML model. We transform the data before it reaches their model — same classifier, same training, just a richer view of the same sensor readings.

Why it works: the quantum feature map captures how sensors move together (non-linear correlation) — something the raw 4 signals don't show on their own.

1. Metrics glossary — the short version

Use this table as-is. Don't improvise these definitions.

Term	What it means
Recall	Of all the real ruptures, how many the model caught.
Precision	Of all the alerts the model raised, how many were real ruptures.
Average Precision (AP)	How well the model tells real ruptures apart from healthy cases, across every alert threshold at once. Higher = better separation.
Average Precision Gain	How much better one representation separates ruptures from healthy cases than another.
Precision @ 90% Recall	Fix recall at 90% (both models catch the same 90% of ruptures) and compare precision at that one point.
AUC	The chance the model ranks a real rupture above a healthy case, picked at random. 0.5 = random guessing, 1.0 = perfect.
False Alarms (%)	Of all the healthy cases, how many still triggered an alert.
Correlation / non-linear coupling	How much the features move together. Low = they don't; high = they do.
p-value	How likely this result is due to chance. Very small = not chance.

2. Walking the screen

2.1 Source Data (the machine + 4 sensors)

Animated shovel diagram + 4 sensor cards.

Sensor	Unit	Plain meaning
Hoist Load	kN	How heavy the load on the cable is, right now
Crowd Vib.	mm/s	How much the digging arm is shaking
Drive Temp.	°C	How hot the motor is running
Cable Tension	MPa	How much strain the cable is under

Say: these four readings are what the model watches to predict a rupture. Same four numbers feed everything else you're about to see.

2.2 "Run DQFE Engine" — the Kipu benchmark reveal

What's on screen: a progress animation, then 3 KPI cards + real plots from Kipu Quantum Hub.

Say once, about the animation: these results were already computed by Kipu's quantum service, on 3,000 real mining shovel records. The animation is revealing them, not computing them live.

Walk the 3 KPI cards in this order — top to bottom, left to right:

Step 1 — Average Precision Gain: +38.3% Compares two representations on the same 15 test splits: **raw** (4 sensors) vs. **hybrid** (4 sensors + 7 quantum-derived features, given to the model together). Same classifier both times. Statistically confirmed ($p \approx 1$ in a billion).

Define "hybrid" the moment you say it: *"the model sees the raw sensors and the quantum features together."*

Step 2 — Precision @ 90% Recall: 39.0% (hybrid) vs. 26.0% (raw) Both models catch the same 90% of ruptures. At that point, hybrid's alerts are real 39% of the time, raw's 26%. Roughly half as many false alarms for the same protection.

Step 3 — Three-way comparison

Representation	Features	Average Precision
Raw	4 (sensors only)	34.5%
Quantum	7 (quantum only)	68.1%
Hybrid	11 (both)	72.8%

Say: quantum alone already nearly doubles raw. Adding raw on top gets the rest of the gain.

2.3 How to read the plots

Click "**▼ Show Plots & Correlation Matrices**".

Precision-Recall curve — know this one cold:

- X-axis: recall. Y-axis: precision.

- 4 lines, one per representation, each sweeping its alert threshold from strict to loose: raw (orange, lowest), quantum (light blue), selected hybrid (green, a secondary check), hybrid (navy, highest).
- Shaded band around each line = spread across the 15 test folds. Raw's band and hybrid's band barely overlap — that gap is real, not random noise.
- Flat dotted grey line ("prevalence 0.257") = what a random guess would score.
- Vertical line at recall 0.90 = where the Step 2 numbers (39% / 26%) come from.
- Average Precision = one number summarizing the whole curve, not just one point on it.

Quantum Feature Matrix: same kind of correlation grid, for the 7 quantum features, plus a bar chart showing which feature carries the most predictive weight. One line: *"this shows which of the seven quantum features does the most work — and it's an interaction between two sensors, not a direct reading."*

Raw Sensor Matrix: same grid for the 4 raw sensors — near-zero everywhere. One line: *"on their own, the raw sensors barely relate to each other."*

2.4 Classical vs. Quantum side-by-side (2A / 2B)

Left card = 4 raw sensors. Right card = 7 quantum-derived features. Same classifier both sides.

Say: left side is untouched; right side is the same 4 signals passed through the quantum feature map. Only the representation changes.

Heatmap: warm = features rise together, cool = one rises as the other falls. Raw: barely any color. Quantum: real structure.

Scatter plot: each dot is one real reading, orange = deviation, teal = healthy. Quantum space: colors separate. Raw space: they overlap.

2.5 Impact (4-step strip)

Feature Expansion → Non-linear Coupling → AUC Enhancement → **False Alarms Avoided.**

Lead with False Alarms Avoided, not AUC — it answers the real business question: fewer unnecessary shutdowns, same detection coverage.

2.6 Live Board — "quantum catches it, classical doesn't"

The centerpiece scene. Open  **Live Board** →  **Simulate perillas** → click **"Quantum catches it (Classical misses)"**.

Loads Record #36, a real, known rupture:

Sensor	Value
Hoist Load	124.2 kN
Crowd Vib.	6.6 mm/s
Drive Temp.	39.7 °C
Cable Tension	119.3 MPa

Say while flipping the switch: *"Same numbers, same machine. Classical: no alert. Quantum: catches it."*

Do this twice — once in the walkthrough, once again near the close, slower.

3. Anticipated questions

Question	Answer
What does the model predict?	Whether the machine is heading toward a cable rupture, early enough to act.
What is "hybrid"?	The model sees the raw sensors and the quantum features together.
Is the mapping applied to the classical side too?	No — classical is untouched; quantum is the same data, mapped. That's the experiment.
Is this the best possible classical model?	No. A non-linear classical model (gradient boosting) on raw alone reaches AP 0.895 — shown on the chart as a reference point, not part of the tested comparison. The tested question is narrower: does adding quantum features improve a fixed, simple classifier? Yes, confirmed.
Is this a quantum advantage claim?	No. The quantum features are computed from the same 4 sensor readings. The claim is: this transformation measurably helps this classifier, on this data.
Is the dataset real?	The 3,000-record benchmark is a synthetic digital twin. The live interactive demo (2A/2B, Live Board) runs on 1,000 real evaluated records.
How often does a rupture happen, and what does it cost?	See §4.
Can I get access?	Yes — see §5.
Is this running on a real quantum computer right now?	The batch results were computed once by Kipu. What you're driving live is the same trained classifier scoring real cases — not a live quantum circuit.

4. Business figures — not fully confirmed

- **Cost per rupture:** ~\$50,000 (mentioned in the prior call). Confirm before quoting as a hard number.
- **Frequency:** not available. If asked: *"We're gathering the exact frequency from the client's data — what we know today is each event is a costly, unplanned shutdown, and this system catches the early signal before it happens."*

5. Closing

- **Live demo:** <https://quantum-feature-mapping-lab.vercel.app/> — any browser, no install, no login.
- **Run it locally:** repo `QuantumFeatureMappingLab` — `npm install && npm run dev`.

- Offer: "If you want to explore this on your own data, let's talk about what that would take."
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6. Do not say

- The engine "executing now" — say "revealing" precomputed results.
- "14 observables" or a live local circuit — the real, active features are **7**, computed once by Kipu.
- "Quantum advantage" or "beats the best classical model" — it's a fixed, simple-classifier comparison; a stronger non-linear classical model exists and is disclosed on the chart.
- Do use "hybrid" and "+38.3%" — they're real Kipu results and the intended headline. Just define "hybrid" the first time you say it.